

Cycles in competitive electricity markets: a simulation study of the western United States

Andrew Ford*

Program in Environmental Science and Regional Planning, Washington State University, Pullman, WA 99164-4430, USA

Received 17 August 1999

Abstract

This paper describes the potential for power plant construction to appear in waves causing alternating periods of over and under supply of electricity. The end result would be major swings in market prices as the industry moves through the phases of a construction cycle. This paper begins with some background on why these cycles should be taken seriously as we write the rules for a restructured electricity industry. It uses computer simulation to learn that cycles could emerge if the western states adopt the market rules used in California. Construction cycles are a potentially serious problem, but they are not inevitable. This paper uses computer simulation to show that cycles could be dampened substantially by introducing a constant capacity payment along side of the market clearing price for energy. The paper concludes with an examination of the consumer impacts of a constant capacity payment. Wholesale consumers would experience higher costs in the short run, but lower energy prices would nullify the impact of capacity payments in the long run. Retail consumers would not necessarily face higher costs in the short run because of a reduction in charges for recovery of stranded costs. © 1999 Elsevier Science Ltd. All rights reserved.

1. Background

Construction cycles have been observed in a variety of industries including automobile manufacturing, real-estate construction and commodity production (Achi, 1998; Ford, 1999; Meadows, 1970; Sterman, 1999). The cycles have not been as prominent in the regulated electric industry because of the investor-owned utilities (IOUs) clear obligation to build the power plants needed to keep pace with demand. Nevertheless, IOUs have felt the impacts of building too much or too little capacity (Ford, 1997). IOUs responded to the challenge with improved forecasting methods and by shifting investments away from capital intensive, long lead time generating technologies. Large nuclear and coal plants were no longer the top choices. By the 1990s, the IOU's marginal resource was substantially smaller, cheaper, cleaner, and it could be initiated only a few years in advance of

need.¹ With the prospects for modular, short lead time construction, the traditional IOUs could look forward to an era when construction cycles would not be a serious problem.

But the electricity industry will no longer rely on traditional rules. When restructuring is completed, the IOUs will be under no obligation to serve any or all of the demand. We will look to the market to deliver the power plants that will be needed in the future. This paper addresses the challenge of designing a market that can do the job in a stable manner.

¹ The shift to modular construction is particularly dramatic in the resource plans issued by the Northwest Power Planning Council (NPPC). Their 1991 plan describes the long-term marginal resource as a combined cycle generator fueled by gas from a coal gasification plant. The levelized cost was 60 mills/kwh; a typical 420 MW plant would take 7 yr to build and cost over \$1 billion. Just 5 yr later, the NPPC described the region's long-term marginal resource as a gas-fired, combined cycle combustion turbine (CCCT). The levelized cost was only 30 mills/kwh; a typical 228 MW unit would take only 4 yr to build and cost around \$150 million.

* Fax: 001-509-335-7636.

Before focusing on the western US, it is useful to learn from others with more experience in deregulation/restructuring. Appendix A summarizes the experiences in the United Kingdom (UK). It concludes that there is little evidence that the UK is progressing toward a market designed to stimulate steady investments in new generating capacity.

It is also useful to review previous modeling efforts. Appendix B provides a brief review starting with an exemplary application of computer simulation in the UK. Appendix B then turns to modeling activities by the California Energy Commission (CEC) and the Northwest Power Planning Council (NPPC). These organizations have used a variety of models to forecast market clearing energy prices. Unfortunately, their models rely on a “perfect foresight” assumption that precludes the possibility of simulating a construction cycle. Investors do not have perfect foresight; they can only proceed with their own best estimates of future prices. Consequently, NPPC planners warn that their price forecasts

are likely to result in a smoother pattern of future prices than the actual market is likely to achieve. Most commodities tend to go through cycles of over and under supply and electricity is not likely to be different. The real electricity market is likely to experience price cycles of several years duration, in addition to the hourly and seasonal cycles that the model captures quite well.

The NPPC warning arises from a commonly held view that the transition to competitive electricity markets will create an industry with some of the standard problems observed in other commodity industries. It is useful to consider the cycles in other industries because their experiences could shed light on analogous problems in competitive electricity markets.

1.1. Electricity versus other commodities

The construction cycle is a chronic problem for agricultural commodities (Meadows, 1970) and metallic commodities such as aluminium smelting (Ford, 1999, Kaufmann, 1983). The aluminum industry is well known to utility planners because of its huge electricity requirements. It is also well known for the tremendous price volatility shown in Fig. 1. Fig. 1 shows price swings of plus-or-minus 100% in just a few years. Near the end of the 1980s, for example, the price of ingots increased to over 100 cents/pound. Within the next two years, it had fallen to only 50 cents/pound. Another upswing in the late 1980s brought the price to over 100 cents/pound, but it returned to around 60 cents/pound by the end of the decade. It is no exaggeration to call the aluminum industry a “roller coaster” industry (Kaufmann, 1983; Oregonian, 1995).

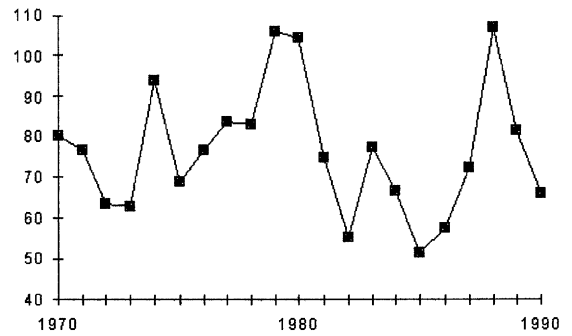


Fig. 1. Price of aluminum ingots in cents/pound, 1987 \$ (Ford, 1999).

Utility planners are concerned that the restructuring of the western electricity markets could lead inadvertently to another “roller coaster” industry. If the analogy with aluminum smelting is extended further, there are concerns beyond the construction cycle. For example:

- Krapels (1996) argues that electricity markets could adopt the “classic” condition of commodities “that capability to supply chronically exceeds typical demand.”
- Newbery (1995, p. 49) makes an analogy between aluminum and the UK electric system. He describes aluminum as an industry “where competition leads to both volatility and periodic price spikes, during which producers recover most of their fixed costs.”
- Backus (1996, p.18) believes that competitors in commodity markets come to prefer highly volatile prices and plan their “tactics” to exploit the volatility for high rewards. He warns us to expect similar tactics in competitive electricity markets.

My discussions with utility planners have revealed a general concern that a construction cycle could emerge in the western US, but there was no consensus on whether the cycles would be more volatile than in other commodity industries. Electricity is fundamentally different from other commodities because it cannot be stored in inventory as a buffer between producers and consumers. Production and consumption must occur simultaneously. This unique requirement leads some planners to reason that an electricity production cycle could be more volatile than the cycles in other commodity industries. This reasoning seems to make sense because a reduction in the size of target inventories can be a destabilizing factor for agricultural commodities (Meadows, 1970). But a contrary view also makes sense. The simultaneity of production and consumption requires the industry to carry an extra reserve of generating capacity. Some planners argue that this extra capacity could serve as a “buffering element” that might be missing from other commodity industries. They reason that an electricity production cycle could turn out to be less severe than cycles observed in other industries.

As we consider these views, we should remember that the requirement for simultaneous production and consumption is just one of several complexities that defy our ability to reason by analogy (Loehr, 1998). Although a construction cycle could be a real possibility, we are unlikely to learn more through analogies with other industries. It makes better sense to apply computer simulation to examine the dynamics of an electricity construction cycle.

1.2. *Organization of the paper*

Section 2 describes a model designed to simulate the potential for cyclical behavior in the electricity markets in the western USA. The model simulates the key feedback mechanisms that give rise to construction cycles. Section 3 presents simulations which reveal that cyclical behavior could emerge under a wide range of assumptions on demand, fuel costs, construction costs and investors' behavior. Section 4 illustrates the use of computer simulation to search for a more stable design. It reviews the arguments for and against capacity payments, and it argues that constant capacity payments deserve closer study. Computer simulations are presented to show the magnitude of a capacity payment that would dampen the cyclical tendencies. The paper concludes by showing "the price of stability" to consumers buying electricity at wholesale and at retail. The concluding analysis reveals that we can achieve more stable electricity markets, and the "price of stability" is surprisingly small.

2. Model description

The model was originally developed for a western utility to help planners think through strategies to achieve cash flow and rate-making objectives. They adopted a long-term perspective that looked several decades into the future, and they were convinced that construction cycles should be taken seriously. These concerns led to the development of a system dynamics² model to simulate the western electricity market along side of the utility's internal considerations. The relevant portion of the model has been removed and adapted for this paper with all parameters estimated from public data. It is designed at a relatively high level

of aggregation compared to the detailed production simulation models described in Appendix B. Rather than simulating electricity demand for 12 different regions, for example, total demand is simulated for the entire region.

2.1. *Demand for electricity*

The demand for electricity in the Western System Co-ordinating Council (WSCC) region is set to 83,000 aMW at the start of the simulation. The user specifies the demand growth rate over time. This paper adopts the standard assumption of 1.5%/yr, the base case growth rate used in the NPPC (1998) study. The growth rate is assumed to apply year after year, thus eliminating variability or cyclicity in demand as a confounding factor. Any cyclicity in the simulations will be traced to the interaction of the market and investors, not to exogenously imposed variations in demand growth.

The model represents the electricity demand as the energy required over the entire year after the impact of demand-side management programs.³ We do not simulate the variations in demand over the hours in a day, over the days in a week or over the seasons of a year. These variations are treated in great detail by the collection of production simulation models used by the CEC and the NPPC. We will take advantage of the short-term forecasts from the more detailed models to provide a check on our results.

The model assumes that the total demand is satisfied by a power pool operating under the power exchange (Px) rules adopted in California.⁴ The Px serves as a central energy clearinghouse to dispatch generation to meet load.⁵ The choice of which generators will be used is accomplished by an auction. Generators submit bids which are presumably based on their short-run variable costs. The Px allows consumers to submit demand side bids with a cap on the price. If the Px cannot clear the market below the cap, their request for electricity will go unserved. Although demand-side bidding is allowed in California, the model assumes that all of the demand will

² System dynamics has its roots in control engineering and has been defined as "that branch of management science which deals with controllability" (Coyle, 1977, p. 2). The method was pioneered by Forrester (1961) and has been put to good use in the regulated electricity industry (Ford, 1997) and the competitive electricity industry (Bunn and Larsen, 1992; Lomi and Larsen, 1999; Vlahos, 1998). System dynamics is especially useful when we need to understand the feedback mechanisms in a system.

³ Demand-side-management programs are falling out of favor as the utilities brace themselves for an era of competition. The current emphasis is on "industry transformation" programs in which utility funds are used to leverage important changes in a national industry such as lighting and motors. Both of these programs are outside the scope of the model.

⁴ This is the approach adopted by the Aurora model described in Appendix B. California has advanced further than other western states in restructuring; its rules are better defined and more amenable to simulation analysis.

⁵ More than 80% of California's electricity has been traded through the Px since it opened in March of 1998, and the exchange is cited as a reference on daily prices for other markets (Moore, 1999).

be bid without a price.⁶ This means that the market clearing price will be set at the cost of the most expensive generating unit needed to meet the total demand. If the region is short of resources, the pool operator will be forced to impose an administrative cap to limit the price. Under these circumstances, the model imposes a “circuit breaker” mechanism to interrupt the normal operations. Rather than allowing the Px Price to climb to excessive levels, we assume managers would use their judgement to limit the Px Price to a high value and hope that new power plants will eventually come on line to eliminate the shortage.⁷

The growth in demand is imposed exogenously by the model user, so the model does not respond to changes in the Px Price over time. Since the level of demand influences the market-clearing price, we have ignored a form of information feedback that was important in previous studies of electric utilities (Ford, 1997). We decided to ignore the link from the Px Price to demand because we did not wish to simulate the step from wholesale to retail

rates and because we feel that retail consumers may well be shielded from changes in the wholesale rates.⁸

2.2. Existing low-cost generation

There are around 2000 generating units in the WSCC, and some models are designed to simulate each and every unit. For our purposes, it makes more sense to combine resources into broad categories like hydro, nuclear and cogeneration. These three categories are assumed to have low variable costs or operating constraints that require them to operate in a “must-run” mode. Nuclear capacity begins the simulation at 9210 MW and the user may specify retirements of nuclear capacity during the course of the simulation. (Retirements are set to zero in the simulations shown in this paper.) The hydro-electric generations is broken down into energy from dams in the northwest and energy from the rest of the WSCC. Northwest hydro-electric generation is set at 15,442 aMW which is representative of the amount expected in a year with average runoff and current operating rules. Hydro-electric generation from the rest of the west is set at 5000 aMW.

This collection of assumptions adopts a highly simplified view of existing generation. Hydro-electric generation, for example, will change from one year to the next due to changes in the weather, possible changes in operating rules and changes in the dams themselves. It is also highly unlikely that nuclear generation will remain constant over time as well.⁹ Surely some change in the output from existing generation is to be expected, so why should we proceed with the assumption that existing units will deliver a constant amount of energy?

We assume constant output from the low-cost generators to maintain focus on the construction cycle. If we

⁶ Once again, we follow the approach used by Aurora. Ignoring demand side bidding is reasonable if one believes that consumers place a high premium on a reliable supply of electricity. (Rose, 1995) argues that customers are willing to “pay at least 100 times the average price” to avoid curtailment of electricity.) On the other hand, certain consumers with huge electricity requirements (like the aluminum smelters) can estimate the maximum price of electricity that would allow them to continue operations in a profitable manner. These industries might turn to demand-side bidding in the Px, or they might negotiate long-term contracts with special interruptible provisions to deal with their unique situation. Simulating these actions is beyond the scope of the current model.

⁷ Administrative limits are a difficult topic since a discussion of how they would be imposed is an admission that the market mechanism may not work effectively. It is not surprising, therefore, that information on price caps is limited. In testimony on the California pool, Grow concludes that “Apparently, the intent is that the market clearing price should be high in this oversubscribed condition and that some of the demand might drop in response to the high price. Perhaps details will be filled in later” (Grow, 1996a, p. 3). The model imposes a circuit breaker mechanism whenever a shortage of energy develops and the Px Price has climbed to a point that is beyond a reasonable value to stimulate generation. The cap is based on the variable cost of the region’s most expensive gas-fired unit or the total, levelized cost of a new CCCT. The model takes the larger of these two costs and adds a user-specified reserve margin. Under base case conditions, the administrative cap would be around 43 \$/mwh. The 43 \$/mwh cap may sound low compared to “price spikes” that have appeared in competitive markets. Price spikes in the California Px, for example, exceed 150 \$/mwh during August and September of 1998 (Moore, 1999, p. 11). But remember that the 43 \$/mwh is an average over the entire year, and this average value may be created by high price caps imposed for a small fraction of the hours in the year. To illustrate, consider Grow’s (1996b) calculations for a simple system with three types of generators and market clearing prices ranging from a low of 17.5 \$/mwh to a high of 40 \$/mwh. The annual average turns out to be 30 \$/mwh. To match the administrative limit adopted in this paper, one could assume that a price cap of around 330 \$/mwh applies for 5% of the hours in a year.

⁸ To simulate price feedback on demand, one must combine the Px Price with the rules adopted in different states to arrive at the retail price to the customer. In some cases (as in Assembly Bill 1890 in California), the rules may turn out to be quite complicated. The complications are the inevitable outcome of lawmakers’ attempts to “provide something for everyone” (Richard and Lavinson, 1996). These rules strive to protect utilities from a low Px Price by adding a competitive transition charge (CTC) to the consumers’ bill. In so doing, the consumer is shielded from the benefit of low rates in the short-run. To protect consumers from high Px rates, on the other hand, the state imposes a variety of targets on retail rates that may be charged in California. The overall effect of these rules, is to shield the retail consumer from major variations in the Px Price. For these reasons, it is reasonable to assume that the growth in demand may be specified independently from variations in the Px Price.

⁹ We might see a reduction in nuclear generation if the owners feel they cannot control operating costs. On the other hand, nuclear owners may be stimulated by the competitive market to improve maintenance procedures. In the process, they might discover what Farney (1997) calls “hidden capacity” or what CERA (1997) calls “capacity creep.”

see waves of construction and periods of over/under supply, for example, we will be able to trace the waves back to their underlying cause without becoming lost in a myriad of “external” factors like changes in the weather.

2.3. Existing, high-cost generation

The final two categories of existing generation are the region’s coal-fired and natural gas-fired generators. The simulations begin with 36,238 MW of coal capacity and 33,110 MW of gas capacity. The user may specify retirements of coal and gas capacity over time, but the retirements are set to zero in this paper. The availability factors and the variable O&M costs are also held constant over time.

All coal and gas generation is controlled by the price established by the power exchange (Px). The Px Price is measured in \$/mwh (same as mills/kwh); it represents an annual average price for electric energy. The Px is organized as if all the western states followed the rules adopted in California. We assume that the owners of coal and gas units bid their energy at variable costs, and the Px operator finds the Px Price to clear the market. The model calculates the fraction of coal capacity in operation by finding the heat rate of a coal-fired unit that is on the margin. It calculates the fraction of gas capacity in operation by finding the heat rate of a gas-fired unit that is on the margin. The user specifies the coal prices in 1998, and they increase over time at the rate of inflation. The user specifies the natural gas prices in 1998 and the real rate of escalation. The model finds the fraction of coal and gas plants in operation based on WSCC and CEC data on heat rates of thermal units across the region.

2.4. The Px price

The Px Price is the price of energy found by simulating the procedures in the California power exchange (Px). The electricity demand imposed on the Px is found by subtracting the “must-run” generation from the total demand in the WSCC. For example, the demand on the Px at the start of the simulation is around 47,000 aMW. The Px turns to the region’s coal and gas plants to meet this demand. With the assumptions adopted in this paper, the Px could obtain over 58,000 aMW if the Px Price were set sufficiently high to bring all of the coal and natural gas units into operation.

The model searches for the Px Price that will bring forth the needed generation through a simple heuristic algorithm. The heuristic approach mimics the Px Price resulting from an auction, but it avoids the type of simultaneous calculations performed by the Px manager. The Px Price is adjusted up or down over time to clear

the market.¹⁰ The simulations begin with the Px Price at around 20 \$/mwh. This price is sufficient to recruit all of the coal capacity and over 60% of the gas capacity into operation during the first year of the simulation. The plausibility of this starting condition was checked by comparison with the more detailed Aurora calculations.¹¹

2.5. Average annual values for demand and price

It is important to remember that the demand for electricity is represented by an annual average expressed in aMW. Consequently, the model does not keep track of the daily or seasonal variations in demand, nor does it simulate the hourly or seasonal adjustments in the Px Price. This means that we must use the annual average price as a single number to summarize a complex distribution of prices over the many hours in a year. Electricity prices in the northwest tend to be higher in the winter than in the summer, and May–June prices are often quite low due to high flows down the Snake and Columbia Rivers. Electricity prices in southwest, on the other hand, tend to be higher in the summer when generation is needed to meet peak loads. Moore (1999) shows substantial variations in the California Px Price within the hours in a single day and concludes that “the price of electricity can change more in one day than the prices of many other commodities change in a year.”

To illustrate how the complex price distribution may be compressed into a single number, consider Grow’s (1996a) illustrative price calculations for the Southern California Edison system. Averaged over 100% of the hours in the year, the marginal cost turned out to be 24 \$/mwh. This single number represents a distribution of marginal costs which range from a low of 12 \$/mwh to a high of 44 \$/mwh. As demand increases over time, Grow expects to see an increase in prices.

After just a few years, the combination of demand growth and closure of older generation may create a seller’s market for energy during peak demand hours and perhaps at other times. In this circumstance, the 205 filing for authorization of the Px provides that the

¹⁰ These adjustments are made quite quickly as the simulation proceeds through time, and the Px Price finds its way almost immediately to the appropriate value to clear the market. This simple heuristic has been tested under a variety of conditions to verify that the model moves extremely quickly to the market clearing price. A sudden reduction in the aMW from hydro, for example, leads to a sudden increase in the Px Price to clear the market. A gradual increase in the demand for electricity, on the other hand, leads to a gradual increase in the Px Price to clear the market.

¹¹ Aurora’s Px Prices are calculated hourly for several delivery points in the WSCC. The hourly prices for delivery at Mid-Columbia were combined into an annual price forecast of around 20 \$/mwh which provides a credibility check on our initial result.

marginal clearing price will be set either at what demanders are willing to pay for energy, or should sufficient information about what they are willing to pay be lacking, at “an administratively pre-determined cap”. The filing does not give any clues as to how high the cap might be, but preliminary drafts suggested that it could be 1 or 2 \$/kwh.

(Grow, 1996a, p. 14)

He then shows how a price cap of 1000 \$/mwh, if in effect for only 1% of the hours in a year, could raise the average annual price to 34 \$/mwh, a value which is more likely to stimulate investment in new CCCTs¹².

Grow's results are emphasized here to make a point about interpreting the average annual Px Price in the model of construction cycles. The Px Price is a single number which summarizes the average annual effect of a complex distribution of prices. To summarize Grow's calculation, for example, the Px Price would be just under 30 \$/mwh. To summarize the second calculation, the Px Price would be just over 35 \$/mwh. We rely on a single Px Price because it is believed that the average value of the market clearing prices over the entire year is likely to be the key variable studied by investors contemplating the construction of new generating capacity.

2.6. Investment in new CCCTs

Investors may select from technologies ranging from small-scale-distributed technologies to large-scale coal and nuclear plants. The large-scale technologies are unlikely to draw investors' interest (Schuler, 1996), and distributed generation may not be able to complete. Investors will probably be drawn to combined-cycle co-generation or combined cycle, depending on the price of natural gas (Schuler, 1996). CCCTs would appear to be the most popular technology for “merchant plants” that are constructed to serve the open market (Moore, 1999). This paper focuses on CCCTs with the following attributes:

Permitting interval	12 months.
Construction interval	12 months.

Construction cost	650 \$/kw ('97\$).
Average heat rate	6842 BTU/kwh.
Fixed O&M costs	2.87 mills/kwh.
Availability factor	90%.

With a fixed charge rate of 15%/yr, the levelized fixed cost would be around 16 \$/mwh. If natural gas is priced at 1.50 \$ per million BTU at the start of the simulation, the CCCT's variable cost would be only 10 \$/mw, and the total levelized cost at the start of the simulation is around 26 \$/mwh (in '97 \$). The total cost will increase over time with increases in the price of natural gas. If gas prices reach 2.20 \$ per million BTU by the year 2003, for example, the total levelized cost will climb to around 27.5\$/mwh (in '97 \$). At this level, it matches the long term market clearing price adopted in the CEC forecast (Klein, 1997, p. 7).

We assume that new CCCTs will be constructed based on investors' forecasts of the future Px Price. If investors look into the future and “see” a price exceeding 27.5 \$/mwh, for example, they will apply for construction permits. Permitting delays can range from 5 to 24 months (Thurston, 1999) depending on the state's procedures and the site's location. California has a two-step process for approving plants — a notice of intentions followed by an application for certification, so the permitting delay can be longer than in other states. It can take as long as 24 months to complete both steps, but exemptions may speed the process. We assume that permits are granted after a 12 month delay.

The model assumes that investors will not apply for a construction permit until they “see” a profitable Px Price in the future. The model does not simulate the unusually large number of applications that have appeared in recent years. In New England, for example, the number of proposed plants was “equal to, if not twice, the installed capacity of the region.” One commentator looked at the 30,000 MW of proposals in New England and estimated that at most 5000 MW would actually move to construction (Thurston, 1999, p. 17) The large number of proposals moving through the permitting process may be caused by investors acquiring pre-approved sites so they will be able to respond quickly when CCCTs eventually become profitable. The model does not simulate investors' attempts to build up a stock of pre-approved sites. This behavior may be simulated implicitly, however, by running the model with the permitting delay reduced to a minimal interval (i.e. 2 weeks).

For the base case simulation, we assume that construction may not begin until 12 months after investor's apply for a permit. The model assumes that investors are continuously monitoring the Px Price. If they believe that profitability has dropped during the 12 month permitting interval, they may cancel the project and there is no construction. But if CCCTs still appear profitable, investors will initiate construction, and the new capacity will

¹² Grow's purpose is not to advocate Px Prices in the range of 1000 \$/mwh to stimulate investment. He elaborates on his purpose in a separate article (Grow, 1996b) which describes a simplified system comprised of peakers, intermediate and base load units. All generation is satisfied; the average annual price turns out to be just under 30 \$/mwh; but the generators net operating revenue is not sufficient to cover their fixed costs. He then calculates the value of a demand-side bid which, if allowed to clear the market for 5% of the hours, will make all generators whole. He concludes that “the solution is easy, because the required bid equals the full revenue requirement of a peaking unit that runs only 5% of the year.” In this case, a demand side bid of 168 \$/mwh which sets the price for 5% of the hours “will make all generators whole.”

come on line 12 months later. With a variable cost of around 10 \$/mwh, it makes sense to combine the new CCCT capacity with the other low-cost units (like hydro and nuclear) and operate them at full availability.

2.7. Investors' behavior

The model simulates investors' expectations for the future value of the Px Price as an internal variable. This is one of the more speculative parts of the model, and discussions with utility planners have revealed a consensus on only two points. First, planners agree that investors do not have perfect foresight, so we do not rely on an optimization algorithm like the one used in Aurora. Second, planners generally agree that investors will look beyond today's Px Price in judging profitability, as advised by Bill Thompson, a senior partner for Black & Veatch:

Builders of new technology had better remember that they'll be competing at tomorrow's prices, not today's. Thompson points to gas plants his firm built in the Northwest: 'Four cents a kilowatt-hour, and now they're shut down ... They're not competitive any more. And they were just finished.

(Schuler, 1996, p. 31)

We assume that investors will observe the Px Price over time to judge the overall trend. The length of this observation interval is set to 24 months. The observed growth rate is then trended into the future to estimate the Px Price that could apply when a new unit would start operation. The investors' "forecasting horizon" is taken as the sum of the permitting interval (12 months), the construction interval (12 months) and an additional interval to look into the first few years of the plant's operating life. For the simulations shown in this paper, the "additional interval" is set to zero.

To illustrate, imagine that the Px Price has reached 23 \$/mwh by the year 2001 and investors have observed an upward trend of just over 7%/yr over the previous interval from 1999 to 2001. Investors extrapolate this trend two years into the future to anticipate the Px Price when a new CCCT would begin operation. The forecasted Px Price would turn out to be just over 26 \$/mwh. If this exceeds the total levelized cost of a CCCT, investors proceed with applications for the construction permits. If the Px Price continues on an upward trend into the year 2002, the forecasted Px Price will be even higher when the construction permit is granted. Under these conditions, investors commit to construction, and the new capacity comes on line in the year 2003.

This example illustrates the timing of capacity initiations, but what about the magnitude of capacity initiations? This part of the model is also highly speculative, and my discussions with utility planners reveal a wide range of opinions. It is generally agreed, however, that

initiations would increase with increases in perceived profitability. Higher profit ratios mean more initiations; lower profit ratios mean fewer initiations. Another point of agreement is that investors do not act as a group with an overarching goal of building exactly the capacity to keep pace with growth in demand. Capacity initiations could turn out to be well below or well above the initiations that would be needed to keep pace with growth in demand. In their regular *Electric Power Watch*, CERA consultants warn that

developers may add substantial speculative capacity So that an oversupply could emerge As we see it, there are many players poised to invest in supply. They need only modest encouragement, and modest encouragement alone will probably produce an oversupply and dashed price expectations.

Such an outcome could discourage supply investment the next time conditions become tight (CERA, 1997). A final point of agreement among planners is that initiations would be limited by the manufactures' capability. The simulations shown in this paper assume that their ability to deliver CCCTs into the western market is far higher than the rate of growth in electricity demand. The model sets capacity initiations as proportional to the investors' perceived profitability ratio from comparing their forecast of the Px Price with the full, levelized cost of a new CCCT.¹³

2.8. Feedback mechanisms

The focus of this paper is the stability of construction and Px Prices over time, so it is important to appreciate the principal feedback mechanisms in the model. Fig. 2 shows the two balancing loops which will dominate the simulated system over time. The inner loop is highlighted with dark arrows. It represents the rapid adjustments in generation from existing power plants responding to changes in the Px Price. This loop acts without delay to keep the supply and demand on the Px in balance. (The "seesaw" in the middle of the loop reminds us of the balancing function of the loop.) This

¹³ The proportionality constant, alpha, is a highly uncertain parameter which is set at 2 for a base case simulation. If investors' profitability is at 20%, for example, initiations are set at 40% of the initiations that would be needed to keep pace with demand. If profitability increases to 100%, initiations are set at 200% of the initiations needed to keep pace with demand. Alpha represents a simple guess as to investors' responsiveness to perceived profitability. It would be desirable to estimate this parameter from data on permit applications, but this estimate would be clouded by the apparent tendency for investors to stock up on pre-approved permits well in advance of profitability. Since we can only guess at the size of this parameter, it's important to remember that its value is highly uncertain and should be the subject of sensitivity analysis.

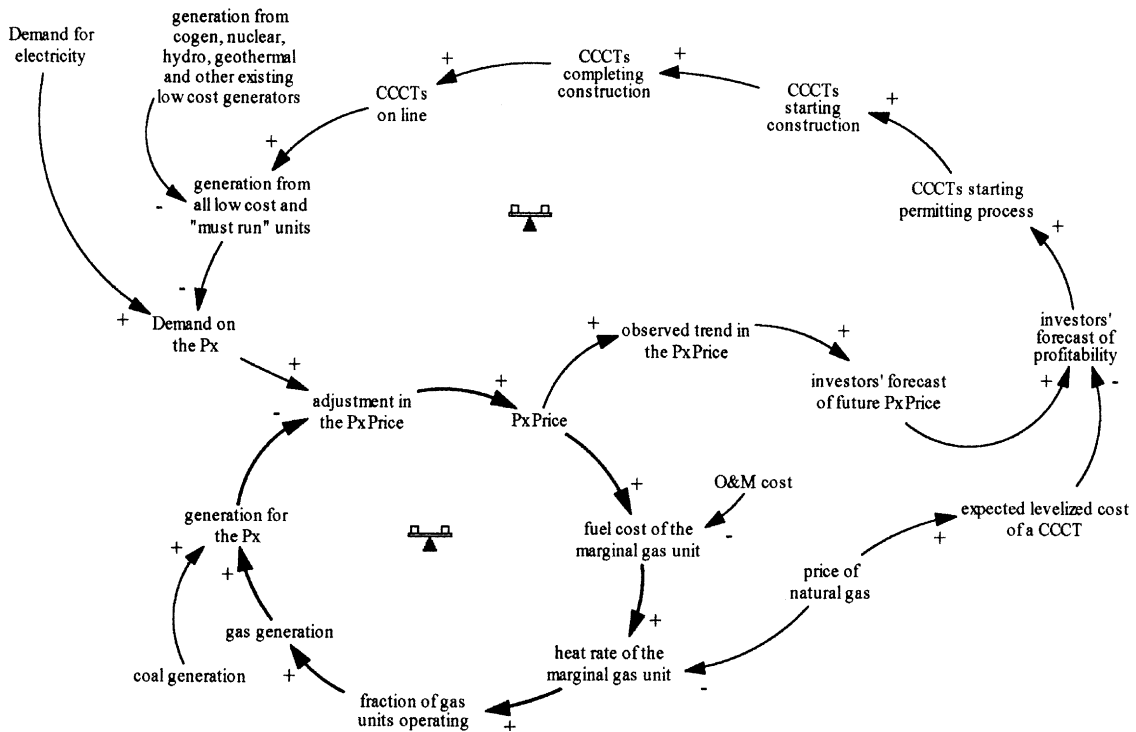


Fig. 2. Two balancing loops which will determine the stability of the Px Price.

loop can do its job as long as there is sufficient generating capacity in the region.

The upper portion of Fig. 2 shows the collection of assumptions on investors' behavior. To trace the cause and effect around the outer loop, suppose that growth in demand leads to higher Px Prices over time. The increase in Px Price leads to an increase in investors' forecast of future Px Price, an increase in investors' forecast of profitability, and an increase in the number of CCCTs starting the permitting process. After a 12 month delay, there will be an increase in the number of CCCTs starting construction, and after another 12 month delay, an increase in the number of CCCTs coming on line. The increase in capacity will reduce the demand imposed on the Px and lead to a downward adjustment in the Px Price.

This example reveals that the closed chain of cause and effect acts to counter the increase in the Px Price. Once again, we have negative feedback, and Fig. 2 includes a "seesaw" to remind us of the balancing function of the outer loop.

The feedback mechanisms in Fig. 2 may prove sufficient to maintain a relatively stable pattern of prices over time. But there is no guarantee that this system will be stable just because we have negative feedback. It is certainly possible that delays in the actions of the construction loop could lead to periods of under and over supply. To sort out the possible patterns, we turn to computer simulation.

3. Simulation results

Fig. 3 shows two variables from the first 8 years of a simulation with base case assumptions. Both the Px Price and the levelized cost of a new CCCT are measured in \$/mwh, in constant 1997 dollars. The Px Price begins the simulation at 20 \$/mwh, around the same result obtained by the Aurora model in the NPPC (1998) study. The demand for electricity grows at 1.5%/yr, so the Px manager is forced to increase the price to bring more and more gas units into operation. The price of natural gas escalates at 1.0%/yr. Escalation in the gas price contributes to the upward trend in both the Px Price and the levelized cost of a new CCCT. By the end of 8 years, the Px Price has increased to the point where the power exchange is clearing at exactly the leveled cost of a new CCCT.

The pattern in Fig. 3 is typical of the short-term results from the Aurora model described in Appendix B. When the Aurora calculations are extended beyond the short term, the Px Price remains exactly at the levelized cost of a new CCCT. This is a plausible result which might appear if new CCCT capacity comes on line in exactly the right amount to counter the growth in demand. But in a competitive market, investors are not striving to bring exactly the right amount of capacity on line. Their goal is to invest in a profitable manner, and we need to extend the simulation to learn if their investments result in a stable price trajectory.

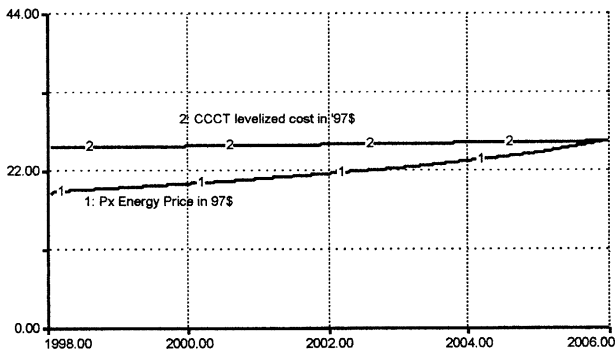


Fig. 3. Short-term simulation results: the Px Price increases gradually over time due to growth in demand, eventually reaching the cost of a new CCCT.

Fig. 4 extends the simulation to reveal that investor activity would appear in waves causing cyclical variations in the Px Price. The graph shows the Px Price and the CCCT cost scaled from 0 to 44 \$/mwh. CCCTs in the permitting process, in construction and in operation are scaled from 0 to 40,000 MW. The first wave of investor activity appears around the years 2004–2009. Capacity in permitting reaches a peak of around 10,000 MW in the year 2008; capacity in construction reaches a peak one year later. Installed CCCT capacity begins to grow around the year 2007. By this time, however, the Px Price has climbed to 32 \$/mwh, well above the cost of a CCCT. Installed capacity is growing rapidly by the year 2008, and the new units allow the Px Price to fall from the peak. By the year 2009, the Px Price has returned to the cost of a new CCCT and is on a downward path. The Px Price reaches a low of 22 \$/mwh before returning to a gradual upward trend. The upward trend is similar to the upward trend at the start of the simulation; it is caused by the gradual growth in demand.

By the end of the first wave of activity, around 14,000 MW of CCCTs are in operation, but there are no CCCTs in the permitting process. Investors are reluctant to get involved in new projects until the Px Price approaches profitable values. This does not occur again until around the year 2013. A second wave of activity appears near the end of the simulation when the Px Price peaks at 37 \$/mwh. The second wave of CCCTs comes on line around the year 2017 and drives the Px Price down to 23 \$/mwh by the end of the simulation.

The base case simulation reveals an inherently unstable system — not only do we see oscillations in the Px Price, but the oscillations grow larger over time. These oscillations appear despite a long list of assumptions which create an unusually constant environment. (Electricity demand grows at a smooth, steady rate year after year, and the region's hydro-electric generation remains constant year after year. Moreover, all of the region's thermal units remain in operation, and their availability factors are constant throughout the 20 year time period.)

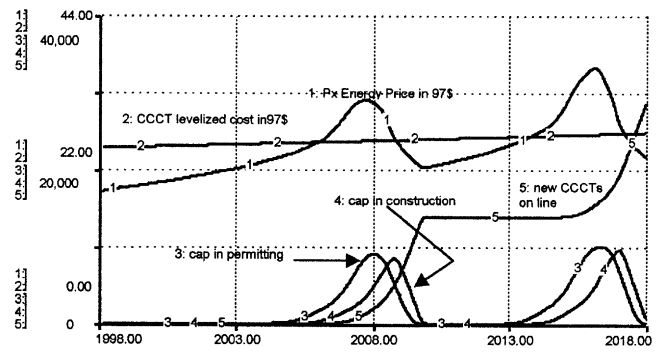


Fig. 4. Base case simulation: construction appears in waves, and the Px Price exhibits growing oscillations over time.

With all of these factors held constant, we know the oscillations come from the interactions inside the system.

Oscillatory systems may exhibit widely varying patterns of behavior. At one extreme, we may see “overdamped” behavior in which the oscillations disappear completely. At the opposite extreme, the same system may show oscillations that grow rapidly into a “limit cycle” (Ford, 1999). These qualitatively different patterns can arise from the same interacting factors operating with different parameters. To sort through the possible patterns, we repeat the simulations with variations in the model parameters.

3.1. The demand growth rate

The NPPC (1998) study suggests that the rate of growth in electricity demand could range from a low of 0.5%/yr to a high of 2.5%/yr. Repeating the base case simulation with the growth rate set at 0.5%/yr shows a gradual growth in the Px Price due to the slow growth in demand. It is not until the very end of the simulation that we would see a Px Price sufficient to interest investors. There are no signs of a construction cycle under these low growth conditions.

The more interesting case involves demand growth rates at the high end of the range expected by the NPPC. Fig. 5 shows the Px Price that appears in a simulation with 2.5%/yr demand growth. The cost of a new CCCT follow the same upward trajectory as in base case. The Px Price, on the other hand, follows an entirely different trajectory. It begins the simulation at 20\$/mwh, the same as before. It then grows rapidly during the early years of the simulation. The price reaches the cost of a CCCT in just 5 years and “shoots past” the cost of a CCCT to reach an administrative limit by the year 2004. The circuit breaker is imposed at this point because the region is short of energy.¹⁴ The price cap would remain in effect

¹⁴ The price cap is based on the cost of the most expensive generating unit plus an administrative reserve. Since the price of natural gas is increasing over time, the price cap increases as well.

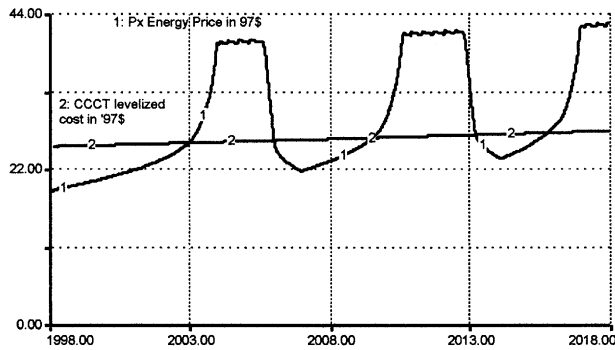


Fig. 5. Simulated Px Price with 2.5%/yr demand growth. The oscillations take the form of a limit cycle.

for over a year before sufficient CCCT capacity comes on line to eliminate the shortage. The new capacity causes the price to fall rapidly, reaching a low of around 22\$/mwh around 2006.

Fig. 5 shows the same pattern reoccurring after the year 2006. Continued growth in electricity demand causes the Px Price to return to an upward trend, and the trend accelerates around the year 2010. The Px Price encounters another administrative limit around the year 2012, and the price is capped until another wave of construction eliminates the shortage of energy.

The oscillation in Fig. 5 is a limit cycle. Limit cycles are common in natural systems (Ford, 1999). They appear when a system prone to growing oscillations encounters a natural limit. Once the system reaches its natural limit, the cycle can repeat itself in an amazingly stable manner (as in the beating of the human heart). But the limit cycle in Fig. 5 is not particularly “natural” since the price caps are imposed administratively on a market that has failed to stimulate the needed investment in a timely manner. It is highly doubtful that the industry could survive the pattern of behavior shown in Fig. 5. It is important that we understand the factors contributing to the unstable behavior and that we find a way to stabilize the system.

One way to learn more is to compare a series of simulations with variations in each of the uncertain parameters. Sensitivity analysis may be performed in comprehensive manner using formal sampling methods (Ford, 1990), or it may be performed informally looking “at one parameter at a time.” This paper follows the informal approach. We set all parameters at their base case values and allow one parameter to vary across a reasonable range of uncertainty. Fig. 6 shows the first test. It compares the Px Prices in simulations with the demand growth rate varying across the range of uncertainty in the NPPC (1998) study.

The first two runs are the slow growth simulations. The first run shows no oscillations for reasons explained previously. The second run reveals oscillations, but the swings in the Px Price are not as large as in the

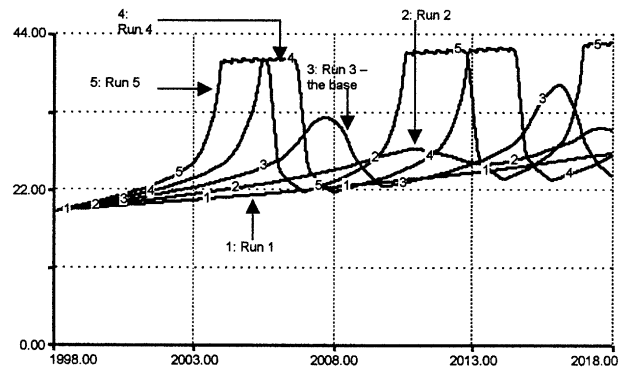


Fig. 6. Px Price with variations in the demand growth rate. Low growth: Run 1: 0.5%/yr; Run 2: 1.0%/yr; Base case: Run 3: 1.5%/yr; High growth: Run 4: 2.0%/yr; Run 5: 2.5%/yr.

base case. The third run is the base case. We see growing oscillations, but the system is never “pushed to the limit.” The fourth and fifth runs show limit cycles. Both simulations employ the same rules for setting the administrative limit, so the prices are limited by the same price cap. The limit is reached somewhat sooner in the fifth run, the simulation with the highest growth in demand.

Two conclusions may be drawn from Fig. 6. First, rapid growth in demand can advance the “long-term” problems closer to the present. With 2.5%/yr growth, for example, the Px Price could be fixed at the administrative limit as soon as the year 2004. With 0.5%/yr growth, on the other hand, the problems of a construction cycle are postponed beyond the 20 year time horizon. A second conclusion involves the simulated volatility. It appears that faster growth leads to increased volatility. This is most apparent by comparing Runs 2, 3 and 4. All three runs show cyclical behavior, and we attribute the cycles to investors’ delayed response to higher prices. A faster growth in demand increases the volatility because the delays in the investors’ response are more problematical when demand grows more rapidly.¹⁵

3.2. The CCCTs

Fig. 7 summarizes a sensitivity test to reveal the importance of the construction cost of a CCCT. Runs 1 and 2 are the low-cost simulations. They show oscillatory behavior, but the swings in the Px Price are not as large as in the base case. Run 3 is the base case with growing oscillations shown previously. Runs 4 and 5 are the high cost simulations. Both show that the system would be “pushed to the limit,” with the first imposition of a price

¹⁵ The destabilizing effect of faster growth is explained in text book examples of logistic models with a lag in the negative feedback (Hastings, 1997; Ford, 1999).

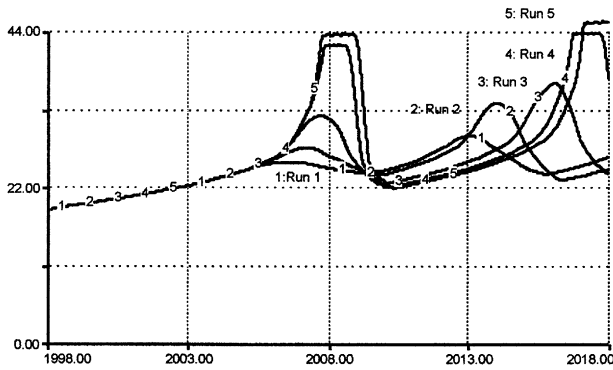


Fig. 7. Px Price with variations in the construction cost of a new CCCT. Lower cost: Run 1: 550 \$/kw; Run 2: 600 \$/kw; Base case: Run 3: 650 \$/kw; Higher cost: Run 4: 700 \$/kw; Run 5: 750 \$/kw.

cap in 2008. The price cap is somewhat higher in run 5 because of the higher cost of a CCCT.

Two conclusions may be drawn from Fig. 7. First, the construction cost has no impact on the Px Price in the short term. All five simulation show the identical Px Price during the first five or six years of the simulation. The second conclusion involves volatility. It appears that greater volatility is to be expected with higher construction cost. For example, we see a limit cycle in both of the simulations with higher construction costs. The Px Price is pegged at an administrative limit for around a year before the new CCCTs come on line to eliminate the energy shortage. When the CCCTs do come on line, they send the Px Price falling to lower levels, and the price remains at depressed levels over a longer interval.

The increased volatility with higher construction costs arises from the incentives needed to trigger an investment. If CCCTs cost more to construct, investors wait longer as they watch for the Px Price to climb higher. The Px manager, in turn, is forced to continue raising the Px Price to bring the most expensive gas units into operation. Eventually, the system is “pushed to the limit” — the Px manager may succeed in bringing every unit into operation, but the region is still short of energy. The system must then operate without a cushion of idle generating capacity, and the Px Price is controlled by administrative limits. This unfortunate situation would not be eliminated until new CCCTs come on line to erase the energy shortage.

The next test turns our attention to the length of the CCCT construction interval. We fix the permitting interval at one year and allow the construction interval to vary from 0.5 to 1.5 yr. The construction cost is held constant at 650 \$/kw, the same as in the base case simulation.¹⁶ Fig. 8 shows oscillatory prices, and the oscillations grow larger over time. Judging from the peak Px

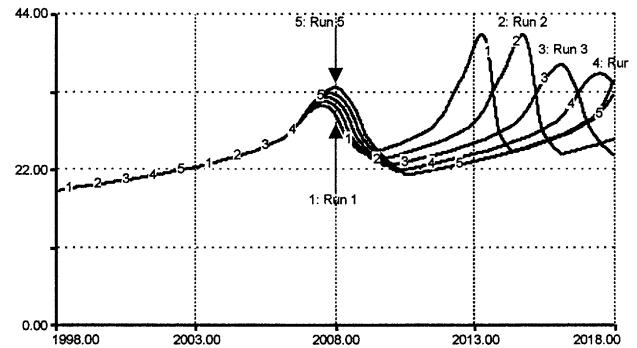


Fig. 8. Px Price with variations in the construction interval. Shorter interval: Run 1: 0.5 years; Run 2: 0.75 years; Base case: Run 3: 1.0 years; Longer interval: Run 4: 1.25 years; Run 5: 1.50 years.

Price around the year 2008, the volatility is greatest with the longer construction interval. Longer construction intervals also contribute to a longer period in the oscillations.

The results in Fig. 8 make sense when we think of the feedback mechanisms in Fig. 2. As a general rule, a longer lag in the response of the construction loop would tend to cause greater volatility (Ford, 1999). But the simulations show only a moderate increase in volatility even though the construction interval varies by three-fold. To interpret these results, we should remember that investors will be aware of the length of the construction interval when they apply for the construction.¹⁷ The model automatically adjusts for the longer construction interval with a longer forecasting horizon. In the fifth run, for example, the investors’ forecasting horizon looks 2.5 yr into the future (1 yr for permitting plus 1.5 yr for construction.) This adjustment partially mitigates the greater volatility that would be expected with longer delays around the construction loop.

3.3. The price of natural gas

Natural gas is burned by existing units which set the Px Price in the near-term and by the CCCTs that are needed in the long term. The price of gas (measured in constant '97 dollars) will escalate at 1%/yr in the base case. Fig. 9 summarizes a test of the importance of the escalation rate. The first run assumes that the price of gas will remain constant over the entire simulation. At the opposite extreme, the fifth run assumes 2%/yr real escalation throughout the simulation.

Each of the simulations starts out with the same price of natural gas, so the Px Price in 1998 is the same in all five simulations. Fig. 9 confirms what we would expect in

¹⁶ The model does not calculate carrying costs during the construction interval. The construction cost and the length of the construction interval are both specified by the user.

¹⁷ The sensitivity tests focus on parameters that are uncertain today. But our uncertainty about the length of the construction interval today does not mean that investors won’t have good information when it comes time to invest.

the short term — higher escalation in gas prices leads to higher escalation in the Px Price. But the long term results may be a surprise. They indicate that the construction cycle is more volatile in a future with low escalation in gas prices. Runs 1 and 2, the two simulations with low rates of escalation, show the limit cycle behavior. They both show that the Px manager would be forced to impose a price cap in the year 2008. Runs 4 and 5, the simulations with higher rates of escalation, both show oscillations, but the oscillations are not as severe as in the base case. Higher escalation in gas prices tends to make the system more stable.

The test of gas escalation in Fig. 9 is similar to the test of CCCT construction costs in Fig. 7. The previous test leads us to expect greater volatility when the Px manager is forced to push the Px Price “to the limit” to clear the market. This unfortunate situation arises when investors wait longer before committing to new CCCTs. In Fig. 7 (higher construction costs) the investors wait longer because they need to see the Px Price rise sufficiently to cover the costs of a more expensive CCCT. The Px manager, in turn, is forced to call upon every gas-fired generator in the region, and the region is still short of energy. A similar situation could occur with low escalation

in gas prices. In scenarios with lower gas prices, the new CCCTs enjoy less of an operating cost advantage over the older, inefficient gas-fired units that will set the market clearing price. The short-term results in Fig. 9 show the market clearing at a lower Px Price when gas is cheaper. This price is not sufficient to trigger investor activity, and the Px Manager is once again “forced to the limit.” Every last generating unit is brought into operation, and the system is operating without a cushion. Administrative limits take effect around the year 2008, and they remain in effect until some new CCCTs come on line to relieve the shortage of energy.

3.4. The investor's behavior

One of the uncertainties in simulating investors' behavior is the magnitude of their response to the forecasted profitability. We assume that capacity initiations are proportional to profitability, and the proportionality constant, alpha, is set at 2. To test the importance of this highly uncertain parameter, five simulations were performed with alpha ranging from 1 to 5. The first run assumes that investors' response to profitability is half as great as in the base case. The fifth run assumes that investors' response is two-and-a-half times larger than the base case. All five simulations reveal volatile cycles in the Px Price. Growing oscillations appear in the three simulations with higher values of alpha, but the cycles are not as volatile as in the base case. The most volatile situation appears in the first run (with alpha set to 1). This test suggests that the willingness of investors to respond strongly to profitability tends to lower, but not eliminate the volatile tendencies in the system.

Another source of uncertainty in investors' behavior involves the length of the permitting interval. Investors may be able to reduce the length of the permitting delay by applying for “blanket exemptions” or by building up a stock of pre-approved sites well in advance of the anticipated need for new power plants. Fig. 10 shows four simulations to test the impact of reductions in the length of the permitting delay. The first run is the base case with a 12 month delay. The second run assumes a 6 month delay. This might be achieved if investors qualify for exemptions on some of the state procedures. The third run assumes an even shorter delay; only 3 months are required before construction may begin. The fourth run assumes that the permitting interval is essentially eliminated. This simulation would represent a scenario in which states grant permits without a review process. Alternatively, it could represent a scenario in which investors acquire a large stock of pre-approved permits. The fourth run assumes that only 2–3 weeks are required for investors to confirm the desirability of a preapproved site and to initiate construction.

Fig. 10 shows cyclical prices across a wide range of assumptions about the length of permitting interval. Run

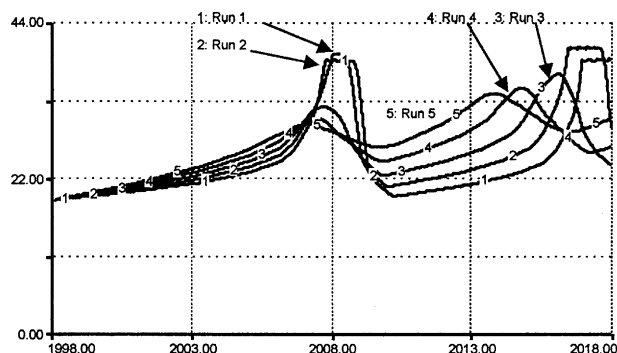


Fig. 9. Px Price with variations in the real escalation in the price of natural gas. Lower escalation: Run 1: 0%/yr; Run 2: 0.5%/yr; Base case: Run 3: 1.0%/yr; Higher escalation: Run 4: 1.5%/yr; Run 5: 2.0%/yr.

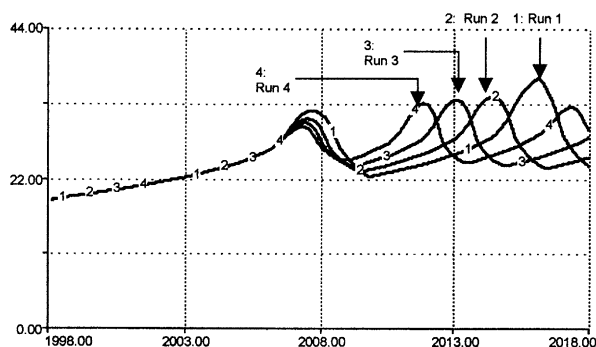


Fig. 10. Px Price with reductions in the permitting interval Base case: Run 1: 12 months; Shorter intervals: Run 2: 6 months; Run 3: 3 months Pre-approved sites: Run 4: 2–3 weeks.

1, the base case shows cycles with the greatest volatility and the longest period. The other simulations show that reducing the permitting interval reduces the overall volatility and the period of the cycles. The fourth simulation yields the most important finding — the system is still prone to cyclical behavior even if the permitting delay is eliminated completely.

4. Design for stability

The simulations in Section 3 shows highly volatile behavior over a wide range of assumptions. We see cycles in the simulated price with different assumptions on investors' behavior, the attributes of new CCCTs, the price of natural gas and the growth in electricity demand. In the more troublesome cases, the oscillations take the form of a limit cycle in which the Px manager is forced to impose price caps during periods of under supply. These cycles appear in simulations with unusually constant conditions. If we see cyclical behavior under these carefully controlled conditions, we must conclude that we are looking at an inherently unstable system.

The simulations envision a western market following the Px rules in California. The California Px is based on the premise that investors will be able to recover the full cost of a new CCCT based on an energy price set in a competitive market. The simulations show that cycles could emerge if we continue to rely on this premise. The cycles are particularly troublesome if CCCT costs and gas prices work against the competitive advantage of a new CCCT. It appears that investors need an additional incentive beyond the Px energy price. At this point, it makes sense to consider capacity payments to provide the needed incentive.

4.1. Background on capacity payments

Hunt (1996, p. 111) explains that capacity payments provide incentives for generators to be available when the system needs generating capacity. They also provide “extra revenue to the generator to cover the capital and other fixed costs which are not covered by the energy price per kwh.” This second role is just what we have in mind in this paper.¹⁸ A variety of arguments have been

advanced on behalf of capacity payments.^{19,20} but many markets have evolved without separate payments for capacity.

Where capacity markets do exist, they tend to be less formal and provide less price reporting than electric energy markets (CERA, 1997). Rose notes that “much has been written about the changing market for economy electric energy. In contrast, prices for wholesale electric generating capacity appear to be hidden.” In 1995, Rose predicted that wholesale markets would respond to a tighter supply/demand balance with prices that “develop a hidden premium over marginal fuel costs.” He calls the hidden premium a “pure capacity price,” and he infers the size of the pure capacity price “hidden” in energy price data from 1997. (The premium prices emerged in summer months in peak demand hours when there was a significant chance of a shortage of generating capacity). Rose asks us to “think of the capacity price as being spread equally over all hours of the year, rather than occurring in certain peak hours, as it does in the real world.” He then calculates the pure capacity price at just over 5 mills/kwh based on the annual capital cost of a new combustion turbine.²¹

4.2. Adding capacity payments to the model

The 5 mills/kwh “capacity price” may be used directly in the simulation model because it spreads the capacity price out over all of the hours in a year. The model has been expanded to permit a total Px Price as the sum of an energy price and a capacity price. The energy price is based on the same rules as before. It's the price needed to bring forth the generation from existing units to meet the demand for energy imposed on the Px. Now, what about the capacity price? Perhaps we should allow the capacity price to vary over time due to variations in the need for generating capacity?

¹⁸ Dickens (1998) reviews the capacity payments adopted in different markets around the world. He reports that several markets have evolved without separate pricing for capacity. These include the markets in Norway, Victoria, Australia, and New Zealand as well as the California Px. “In those markets without a separate capacity price, the market designers appear to have believed that energy prices alone will provide sufficient incentive for new investment in response to market demand.” Some of these markets have been running for several years, but Dickens cautions that there is not enough experience from these markets to conclude whether the energy prices will be sufficient to stimulate the needed investment.

¹⁹ The most relevant argument is by Michaels (1997) who believes it is unwise to rely solely on energy markets. He acknowledges that an energy market can lead to efficient allocation of energy from existing plants, but his concern is for new plants. He argues that an efficient pattern of new construction would occur only as an “accidental” outcome of an energy market. Energy markets will “display unstable prices and chaotic investment patterns even if, in the long run, investors in generation earn competitive returns on average.” Michaels concludes that “timely investment in economically warranted capacity is as necessary for long-run efficiency as marginal-cost allocation of existing capacity is for short-run efficiency, yet very little attention is being given to capacity planning. If an efficient energy market gives rise to an inefficient capacity market, utilities may become the investors of last resort.”

²⁰ Michaels is not alone in calling for capacity payments (Conkling, 1995; McCullough, 1998 and Rose, 1995, 1997).

²¹ The 5 mills/kWh is based on an annual fixed cost of 40 \$/kw/yr. Rose increases this cost by 18% to cover a reserve requirement and spreads the cost over the 8760 h in a year.

A variable capacity payment reminds us of the problems discussed in the UK. The original UK pool set a capacity payment at (approximately) the product of the loss of load probability (LOLP) and the value of the loss of load (VLL). If we adopt this approach, we face the problems described in Appendix A — the LOLP is extremely sensitive to the system reserve margin and subject to manipulation by companies with horizontal market power. The second problem is the dynamic problem revealed in Appendix B — the UK formula would lead to highly volatile patterns of investment, even if we ignored the problem of market power. This second problem is directly relevant to our search for a more stable system. The simulations in Section 3 show a highly volatile system if we rely on energy prices alone. The last thing we need is additional volatility from a capacity payment that rises and falls with swings in the construction cycle. Investors need additional incentives, not additional volatility. It makes better sense to provide the additional incentive as a capacity payment that remains constant over time. We may use the new simulation model to search for the size of the capacity payment to achieve improved stability.

4.3. Simulated impact of capacity payments

Let us begin with a capacity payment of 1 mill/kwh. (This payment would be five times smaller than the “pure capacity payment” identified by Rose.) The total Px Price

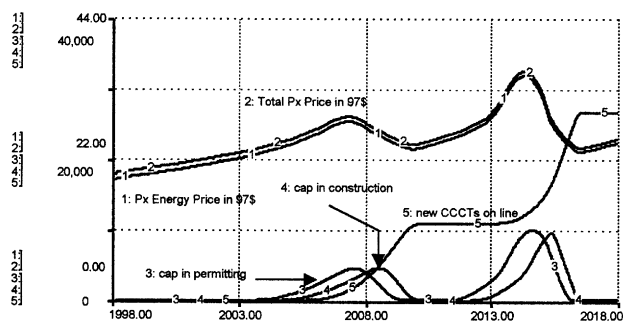


Fig. 11. Base case plus a 1 mill/kwh capacity payment.

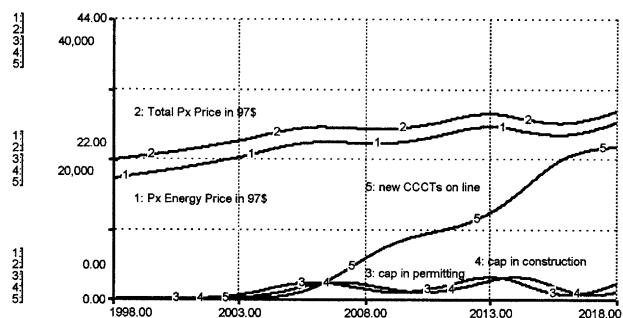


Fig. 12. Base case plus a 3 mill/kwh capacity payment. Construction still appears in waves, but the waves are less volatile.

is the sum of an energy price and the 1 mill/kwh capacity price. Fig. 11 reveals the same cyclical pattern of construction seen before. The first wave of construction appears around the year 2008; the second wave around the year 2015. The waves in construction lead, in turn, to oscillations in the Px energy price and the total Px Price. Fig. 11 shows small, but encouraging changes from the base case in Fig. 5. For example, the first wave of construction starts somewhat earlier. Investors are willing to commit earlier to CCCTs because they forecast future profitability based on the total Px Price (not just the energy price). We see a more gradual growth in installed capacity around the year 2009 and a shorter time interval before investors commit to a second wave of investment.

These differences are small, but they are headed in the right direction. Let us increase the capacity payment to 3 mills/kwh and repeat the simulation, Fig. 12 shows the results; it confirms that increasing the size of the capacity payment leads to greater stability. We still see waves of investor activity, but the cyclical pattern is much less of a problem. The first sign of investor activity appears before the year 2003, and the new CCCTs begin to grow around the year 2004. The growth in new capacity shows only mild variations around the upward trend that is necessary to keep pace with a growing demand. The total Px Price shows only mild swings as the industry moves through the different phases of the construction cycle.

Fig. 13 continues the investigation with a 5 mill/kwh payment, approximately the “pure capacity payment” identified by Rose in the east coast data from 1997. The new simulation shows the stable pattern of construction that we would like to see when demand is growing at a steady rate year after year. Investors begin applying for construction permits around the year 2001, and the CCCT capacity begins to grow by the year 2003. The pattern of growth is smooth and steady because investors fill the “pipeline” by the same amount year after year. The steady growth in new generating capacity allows the energy price to follow a smooth upward trend. The upward trend is caused by a 1%/yr real escalation in the price of natural gas. The total Px Price is 5 mills/kwh

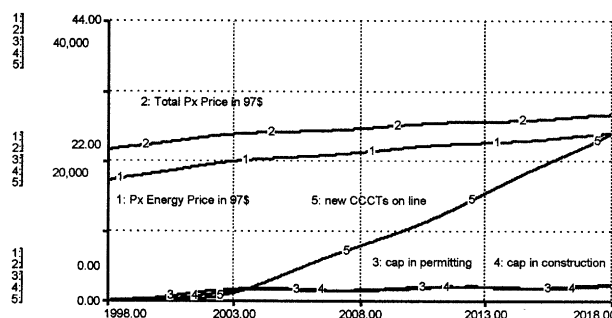


Fig. 13. Base case plus a 5 mill/kwh capacity payment. Construction no longer appears in waves.

above the energy price, so it follows the same upward trend.

4.4. The price of stability

The 5 mill/kwh capacity payment is clearly evident in Fig. 13 because the total price is always 5 mills/kwh higher than the energy price. This may leave you with the impression that the “price of stability” is a constant 5 mill/kwh penalty, year after year. But this would be a false impression. The real “price of stability” is to be found by comparing a simulation relying exclusively on energy prices with a simulation that allows an additional payment for capacity. Fig. 14 provides such a comparison. It shows five simulations with variations in the size of the capacity payment.

The first run has no capacity payment, and it shows the unstable behavior seen previously. The second run shows that a 2 mill/kwh capacity payment would not be sufficient to eliminate the cyclical behavior. Runs 3–5 are more encouraging. They show that the total price would follow a stable trajectory.

To learn the “price of stability,” we should compare Runs 3, 4, or 5 with Run 1, as in Table 1. This comparison reveals that the capacity payment is clearly visible at the start of the simulation. An increase in the size of the payment leads to a corresponding increase in the total price. Run 1, with no capacity payment, has the lowest Px Price. But this advantage disappears over the next 8 years. It is erased by the actions of investors, who see greater profitability from investing in new CCCTs. When the added CCCTs come on line, they allow for the energy demand on the Px to be satisfied with a lower energy price. By the year 2006, the additional CCCTs are sufficient to allow the reduction in the market-clearing energy price to exactly compensate for the capacity payment.

The long-term comparison shows that Runs 3–5 all converge to the same value. (All three simulations shows the total price following a stable trajectory, eventually reaching an equilibrium value of 27 mills/kwh.) This

comparison shows that the “penalty for stability” will be confined to the short term. In this example, the short term is around 8 years, enough time for investors’ response to bring the system into dynamic equilibrium. Once we look beyond 2006, the total price will not be affected by the size of the capacity payment.

Fig. 14 shows that a capacity payment of 4 mills/kwh or higher would be sufficient to eliminate the cyclical tendencies under base case conditions. But we know from the sensitivity studies that the cyclical tendencies are more severe with faster growth in demand and lower escalation in the price of natural gas. To create a more challenging situation, let us increase the demand growth rate from 1%/yr to 2.5%/yr and reduce the real escalation in natural gas prices from 1%/yr to 0%/yr.

Fig. 15 compares the total price with these assumptions and five values of the capacity payment. The first run has no capacity payment, and the model responds with a limit cycle similar to the results shown in Part 3 of the paper. The second and third runs include capacity payments, but the payments are not sufficient to avoid the cyclical behavior. We see limit cycles in the Px Price with a somewhat shorter period and a higher total

Table 1
Price comparison from Fig. 14.

Starting point (1998)	Run 1 shows the lowest price — around 20 mills/kwh Run 3 is 4 mills/kwh higher Run 4 is 6 mills/kwh higher Run 5 is 8 mills/kwh higher
Short term (1998–2006)	Run 1 maintains a price advantage, but the advantage shrinks over time. By the year 2006, the total Px Price is the same in all of the simulations
Long term (2006–2018)	Run 1 shows major swings in the price due to the construction cycle. The average value is around 27 mills/kwh Runs 3–5 avoid the construction cycle. They each arrive at the same value, around 27 mills/kwh.

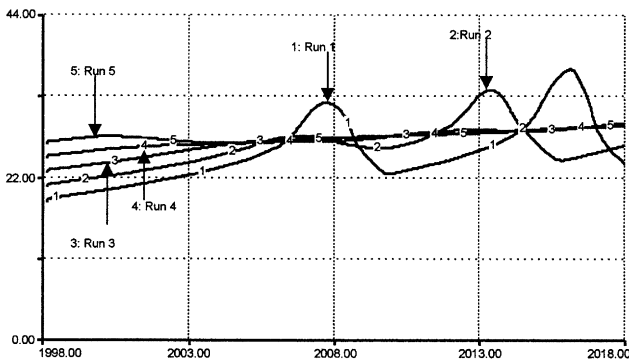


Fig. 14. Total Px Price with variations in the capacity payment. Run 1: 0 mills/kwh; Run 2: 2 mills/kwh; Run 3: 4 mills/kwh; Run 4: 6 mills/kwh; Run 5: 8 mills/kwh.

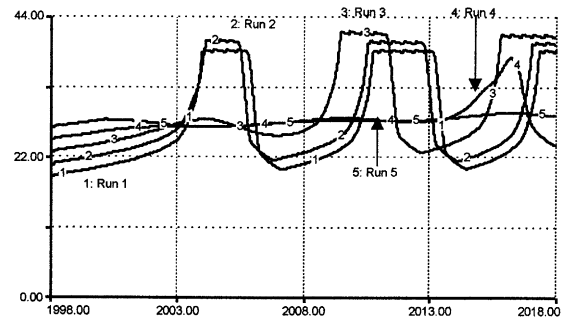


Fig. 15. Total Px Price in simulations with 2.5%/yr demand growth, no escalation in the price of natural gas and variations in the capacity payment. Run 1: 0 mills/kwh; Run 2: 2 mills/kwh; Run 3: 4 mills/kwh; Run 4: 6 mills/kwh; Run 5: 8 mills/kwh.

price when the system is subject to an administrative limit. The fourth run, with a 6 mill/kwh capacity payment, allows for more stable behavior for most of the time period, but the cyclical tendency has not been erased completely. The fifth run is the most stable result. It shows that an 8 mill/kwh capacity payment would allow for a stable, flat trajectory over the entire simulation.

4.5. The wholesale impact of capacity payments

These simulations have shown that a constant capacity payment can convert an inherently unstable system into a stable one. If we want to avoid designing another “roller coaster” industry, it makes sense to add a constant capacity payment to complement the energy price currently used in California.

Now, what is the price to be paid for the stability? With base case conditions, the wholesale impact is evident from Fig. 14 and Table 1. The “price of stability” is paid out in the form of a higher total Px Price over an 8 yr period. A 4 mill/kwh capacity payment is sufficient to eliminate the construction cycle, so the penalty would be 4 mills/kwh in 1998 and decline gradually to zero over an 8 yr period.²²

To interpret the size of the penalty, consider the extra payments to the Px for the energy needed to serve the demand from the three large IOUs in California. Their total energy demand in the year 2000 has been estimated at around 200 billion kwh/yr (ER '96, p. A-45). By “eye-balling” the price differences in Fig. 14, we could estimate the additional wholesale payments at

First Year Penalty in 1998:	Cumulative Penalty over 1998–2006:
4 mills/kwh for 200 billion kwh/yr is	\$400 million per year for 8 yr is
\$800 million	\$3.2billion

Now let us consider the more challenging situation in Fig. 15. To obtain stability under the more difficult conditions, an 8 mill/kwh payment is required. The additional wholesale payments would be:

First Year Penalty in 1998:	Cumulative Penalty over 1998–2003:
8 mills/kwh for 200 billion kwh/yr is	\$800 million per year for 5 yr is
\$1,600 million	\$4.0 billion

²² Rose has argued that energy markets will carry a “hidden capacity price” that will appear in the form of spikes in the energy price during peak periods in a year with tight supply. He might argue that this hidden price would disappear if capacity payments were made explicit — investors would respond with adequate investment in new CCTs, and there would be NO impact on wholesale rates. This paper adopts the more cautious view that wholesale rates will increase in the short term.

These calculations reveal that the wholesale price of stability is around \$4 billion in extra payments to the Px for the electric energy to serve the demand of the three large IOUs in California. The payments would be highest in the years immediately after capacity payments are adopted. They would then phase out automatically over time.

4.6. The retail impact of capacity payments

The \$4 billion penalty could be interpreted as the extra wholesale cost to serve the energy needs of the retail demand of the California IOUs. If wholesale costs increase by \$4 billion, one would normally expect that retail costs would increase in a similar manner. But retail ratemaking in California is governed by the many provisions of AB 1890, the most important of which requires consumers to cover the stranded costs of the three IOUs. The stranded costs have been estimated at around \$26–30 billion (ER '96, p. 20; O'Donnell 1996, p. 33). The IOUs are allowed to recover these costs by adding a competitive transition charge (CTC) to the retail bill. In 1996, the CTC for the three IOUs added \$6.7 billion to the retail bills (ER '96, p. 20).

Fig. 16 places the CTC in perspective by showing charges on a typical bill for a PG&E customer with 600 kwh of usage in 1998. The largest component is distribution charges (36 mills/kwh.) The CTC is the second largest contributor at 34 mills/kwh. The Px energy charge, the focus of this paper, is the third largest charge, at 24 mills/kwh, the energy charge is substantially smaller than the CTC. The remaining charges include 16 mills for the Trust Transfer Account and several smaller charges.²³

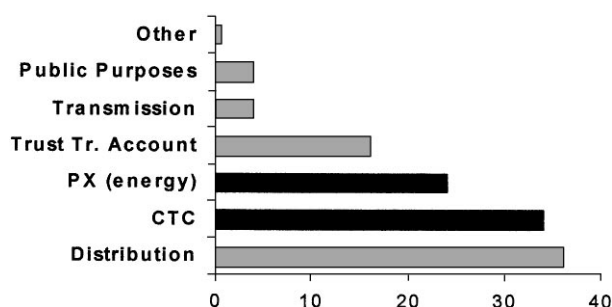


Fig. 16. Typical mills/kwh charges in a California retail bill (CPUC, 1999).

²³ The Trust Transfer Account involves charges on “rate reduction bonds” issued by California’s distribution companies to “make up” for the AB1890 requirement to reduce rates by 10%. The public purpose charges pay for aid to low-income customers, efficiency and renewable programs, and research and development. The remaining small charges pay for nuclear decommissioning and a state regulatory fee.

The Px energy charge and the CTC are shaded in black to draw our attention to these components of the typical bill. These are the charges that will change if capacity payments are adopted in California:

1. The Px energy charge would be replaced with the total Px charge (for both energy and capacity), and the wholesale impact could be around 5–8 mills/kwh immediately after capacity payments are adopted.
2. The CTC charges, on the other hand, would decline by a corresponding amount because the California utilities would earn more from selling to the Px.

The compensating change in the CTC charge arises from the rules of AB 1890 which were written to protect the IOUs from the adverse cash flow of a competitive market. If the market clears at lower prices, for example, the utilities may impose a larger CTC to meet their fixed cost obligations. If the market clears at higher prices, however, a smaller CTC would be required. In this paper, we envision that market prices would clear somewhat higher in the first few years after adoption of a capacity payment. When the utilities earn increased revenues selling to the Px, regulators would lower the CTC. Using the example of the three large IOUs (with 200 billion kwh/yr of demand), the Px charges would increase by around \$4 billion over the next 5 yr. The stranded costs of these utilities are around \$26–30 billion, substantially greater than the \$4 billion. It seems that capacity payments would not penalize California's retail consumers in the short term.

Now, what about the long term? Could capacity payments lead to higher retail bills after the CTC charges are phased out? The California PUC expects CTCs to be completed by the year 2002 (CPUC, 1999, p.7), so one might ask if the retail consumer would experience higher bills after the year 2002. But Fig. 15 reveals that the short-term impacts of capacity payments are largely eliminated by the year 2002. By the start of 2003, for example, the total Px Price is almost identical in all five simulations in Fig. 15. After 2003, the wholesale impacts of capacity payments are beneficial. For example, the total Px Price is considerably lower in the years from 2003–2006. And judging from the overall performance, capacity payments lead to improved stability in rates over the entire time interval.

This discussion focuses on California because California has taken the lead in restructuring and planing for recovery of stranded costs. The California example suggests that capacity payments would lead to a more stable system without penalizing the retail consumer. At this point if time, however, it is hard to extend the California argument to the remainder of the western states. The other states have not yet spelled out their plans in as much detail. Although IOUs in the remainder of the west will face significant stranded costs, they are

not likely to be as large as in California.²⁴ If the remaining states elect to follow the California example, they will turn to CTCs to allow their IOUs to meet their fixed obligations on schedule. If this happens, the retail consumers in the remaining states might also be shielded from the capacity payments described in this paper.

5. Conclusions

Construction cycles are a potentially serious problem if the western electricity markets follow the power exchange rules established in California. Construction of new generating units could appear in waves causing major changes in market prices during different phases of the construction cycle. Under some circumstances, the cycles would take the extreme form of a limit cycle. In this unfortunate situation, the industry would face repeated periods of under supply, and regulators would be forced to intervene with administrative limits on the price of electricity.

These problems arise from the inherently unstable interactions between the power exchange and investors. But the cycles are not inevitable; they could be dampened substantially if western markets allow for a constant capacity price along side of the market clearing price for energy. A capacity payment in the range of 5–8 mills/kwh could stabilize the market in the long run at the expense of slightly higher wholesale rates in the short run. Based on the rate making rules in California, however, retail customers would not feel the impact of the short run increase in wholesale rates.

These conclusions have been drawn from a simulation study based on a pragmatic philosophy that major restructuring "should be based less upon faith in economic ideology and more upon the careful analysis of business simulation models" (Bunn, 1994, p. 367). The long-term stability of electricity markets is an important and complicated topic. It deserves the best efforts of industry planners with the capabilities to expand upon the analysis reported here.²⁵

²⁴ Hirst (1996, p.v) estimates stranded costs for the entire U.S. from \$100 to \$200 billion. He believes stranded costs would be "especially severe in California, New York, Ohio and Pennsylvania" (Baxter and Hirst, 1995, p. 18).

²⁵ The analysis in this paper could be expanded in several directions. First, greater diversity in assumptions (fuel costs, O&M costs) for both existing generators and for potential generators would be a useful addition. (With greater diversity, one might see greater stability.) Another important improvement is to allow existing units to be retired over time. The retirements would probably be concentrated during low points in the price cycle, and they could stabilize or destabilize the system, depending on the magnitude and timing of their response. The model could also be improved by explicit treatment of the region's transmission system and the construction of new transmission capability. And finally, the model could be expanded to treat capacity issues in a more explicit manner. With this improvement, the model could be used to further explore the challenges of adopting the capacity payments recommended in this paper.

Appendix A. lessons from the UK

One of the longest running experiments in restructuring has been underway in England and Wales since the late 1980s. Bunn (1994, p.367) reports that the government's "clear intentions to privatize the UK electricity industry first appeared in the Conservative Party Manifesto of 1987." He describes the initial motivation as highly ideological. It was based on a faith in market forces and deregulated competition.

The UK government dismantled the Central Electricity Generating Board (CEGB) in 1991. The CEGB's fossil fuel generators were sold to two private companies (National Power and PowerGen). The nuclear generators were transferred to a public company. A National Grid Company (NGC) was created to take over the transmission business and to take responsibility for operating a daily power pool. The pool was designed for half-hour prices based on the system marginal price (SMP) associated with the marginal generating unit. In addition, as an incentive to build new capacity, generators received a capacity payment. The capacity payment turned out to be approximately the product of VLL and LOLP, where VLL is the value of loss of load and LOLP is the loss of load probability. The VLL was set by the regulator at over 3 \$/kwh, and the LOLP varied with changes in reserve margin. The intent was to ensure that investors received a timely and adequate signal to promote needed investment. The capacity payments did not amount to much in the first few years of the pool's operation because of high reserve margins. By 1994–1995, however, capacity payments contributed 20% of the generators' earnings, enough to pay for 3000 MW of new capacity, or 6% of the total capacity (Newbery, 1995).

It's been over a decade since the Conservative Party signalled its intent to privatize the UK power sector, but it's still too soon to conclude whether the UK is headed toward a market that will deliver new generating capacity in a timely manner. Their long-running experiment has been beset with a variety of problems, many of which have been attributed to the initial concentration of market power in the hands of National Power and PowerGen (Backus, 1996; Bunn, 1994; Green and Newbery, 1992; Helm, 1994; Joskow, 1997; Hunt, 1996; Newbery, 1995; Wolak, 1996). These companies controlled the key generating units which set the SMP; they also had indirect control over the LOLP which was used to derive the capacity payment. Some observers believe that these companies were able to control the pool prices, either to increase the pool revenues directly or to increase the revenues from CfDs, contracts for differences between the pool price and a contracted price. This concentration of horizontal market power led to a series of complaints, and the regulator responded with a series of price caps. The move to a smoothly functioning market seemed to

be derailed, but what about investment in new generating capacity?

The UK experiment certainly did lead to a major investment — around 10 GW of gas fueled CCCTs were developed in the 1990s. But the new capacity appeared for unusual reasons. The contracts for the CCCTs were signed by distribution companies and independent power producers (IPPs) as a "strategic" reaction by distribution companies looking for protection against the market power of National Power and PowerGen (Bunn, 1994; Newbery, 1995; Patterson, 1997). The new gas generation displaced coal generation and contributed, in part, to a massive closures in coal mines. Coal mine closures have, in turn, amplified the complaints to the government. The government responded with a moratorium on investment in new gas-fired generation through the year 2000 (Thurston, 1999, p. 19). The regulator has responded as well. Besides imposing price caps, he has pressured the large companies into divesting some of their generation, and he has issued a major revision in the "trading arrangements" (Littlechild, 1994; OFFER, 1998). The revision included the capacity payments formula which has been a source of concern to many customers:

Arguments against them are that they are unnecessary; they do not in practice play the theoretical role attributed to them; they are subject to manipulation; and are more likely to increase costs to customers unnecessarily than to protect security of supply. Capacity payments have not induced generators to bid at marginal cost in the Pool: when such payments have been low, generators have simply increased their bid prices or taken other steps to increase SMP to compensate. They have also been able to maintain a similar structure of prices (that is, significantly higher prices at times of higher demand and lower prices with low demand) when capacity payments have been low as well as when they have been high. Generators have also been able to influence the level of capacity payments by closing or "mothballing" capacity. The resulting uncertainty about total prices, and vulnerability to the action of the larger generators, is unlikely to be conducive to new entry.

(OFFER, 1998, p. 152)

The regulator concluded that capacity payments were not necessary and concluded that "it seemed more sensible to focus on revised trading arrangements that did not have the added complication and potential distortion of a capacity mechanism" (OFFER, 1998, p. 39).

With the prohibition on gas-fired generators and the "capacity mechanism" eliminated, one is left to wonder where the UK experiment will head next. The regulator's main priority is to increase the number of competitors to achieve a better approximation to a competitive market. There has been progress in this direction. (Bunn, 1997;

Lomi and Larsen, 1999; Littlechild, 1994; OFFER, 1998). But whether the UK is progressing toward a market designed to stimulate steady investment in new generating capacity is still an open question.

Appendix B. previous simulation studies

This appendix reviews previous simulation studies that are relevant to the study of cyclical behavior in the western USA. Let us begin with a question remaining from the previous appendix — would the UK capacity payment formula have performed well if it were not a convenient lever to be exercised by companies with horizontal market power?

B.1. Simulating the UK capacity payment

This question was addressed by Bunn and Larsen (1992) using computer simulation. Their model calculated the profitability of new power plants and the lead time to bring the plants on line. The simulations indicated that the UK system was inherently unstable — reliance on the capacity payments would lead to cycles of under and over capacity and highly volatile pool prices (Bunn, 1994; Bunn and Larsen, 1992). The volatility was not connected to abuse of the capacity payment formula by PowerGen and National Power, nor was it connected to the “strategic reaction” of the distribution companies. Rather, it was due to a combination of dynamic factors involving the size of the capacity payment, the cost of new generating units, the lead time to bring them on line and limits on investors’ ability to see into the future.

This simulation study provides a useful example of the application of computer simulation to help us learn during a period of rapid change. The modeling can be especially helpful when industry leaders have a paucity of relevant experience (Lomi and Larsen, 1999). The capacity payments formula has been eliminated, so their individual simulations are no longer relevant. But their style of modeling exemplifies a pragmatic approach based on the philosophy that major restructuring “should be based less upon faith in economic ideology and more upon the careful analysis of business simulation models” (Bunn, 1994, p. 367). This pragmatic philosophy makes sense as we turn our attention to the market in the western US. Let us begin by searching for simulation models which might be put to pragmatic use to study volatility in new construction.

B.2. Purpose of computer models

A variety of models are used in the electric industry. Most focus on short-run market analysis dealing with issues of market power, price formation and asset valu-

ation. They employ a variety of techniques including highly detailed production simulation models, statistical models (i.e., models using the Black–Scholes equation) and economic equilibrium models (Kahn, 1998; Ramos, 1998). For this paper, it is important to consider a different category of models. They should look over ten years into the future, and they should be reasonably well documented in a public forum. In the western US, the best place to turn for such models are the organizations with a long history of resource planning, careful documentation and public participation. Two organizations stand out: the California Energy Commission (CEC) and the Northwest Power Planning Council (NPPC).

B.3. The California Energy Commission

The CEC has used a wide variety of models, many in support of the *Electricity Report* proceedings (ER ’90; ER ’96). The CEC relied on a combination of four models to develop a forecast of the market clearing price for the California energy market. (The market began operation in 1998 and is sometimes known as the power exchange or Px.) The four models are explained by Klein (1997). UPLAN (a multi-area chronological production cost model) is the main model for the short-run, the interval from 1998 to 2003. UPLAN was used to simulate the operation of the Western Systems Co-ordinating Council (WSCC) by using approximately 330 power plants and a network equivalent circuit with 15 demand areas, 87 nodes and 219 transmission lines. Other models (such as ELFIN, a single-area probabilistic production cost model) were used to provide supporting information or reference cases. All of the modeling “assumes that bids are based on variable operating costs of the generating units” with an exception for bidders “who under-bid during off-peak hours (to remain on line) and over-bid during on-peak hours to compensate for their off-peak losses.” Klein (1995, p. 8) acknowledges that in a competitive market “there will be some companies whose bidding strategies may be much more complicated than this, but we have not attempted to predict such behavior.” The short-run forecast issued in late 1997 shows the Px Price averaged over all hours in the year and over the entire state of California. Measured in nominal dollars. The forecasted Px Price begins at around 27 \$/mwh in 1998, falls to around 26 \$/mwh in 1999, then climbs steadily to around 29 \$/mwh by the year 2002.

The CEC analysis changes gears after the year 2002. Klein (1995, p. 8) explains that “after 2002 the market clearing price is set outside of the model, to be equal to the price of a new entrant.” He argues that “this is done based on the logic that the market clearing price will rise in response to increased load but once it is high enough to attract new entrants, new entrants will come into the market and drive the market clearing price downward back to its previous level.” These words may convey

a feeling for the ups and downs of a construction cycle, but the CEC forecasters have a different vision of the future. Their 1997 long-term forecast shows the long-run market clearing price at a rather steady value of around 27 \$/mwh. A construction cycle does not appear in the CEC forecast, not does it appear in the forecasting discussion.

The CEC analysts have other issues to discuss. The dominant question in the short run is whether to use average heat rates or incremental heat rates to find the market clearing price. Klein (1997, p. 8) argues that incremental heat rates would “exactly emulate the marginal cost of traditional production cost modeling, and would provide woefully inadequate revenue to survive the market. He estimates that bids based on average heat rates can elevate the annual average value of the market clearing price by as much as 25%. A similar view is given by Kahn (1988) who reminds us that average heat rates are higher than incremental heat rates due to start-up and no-load costs. Kahn argues that a generator submitting bids into a “single-part bidding” pool (like the California pool) will find that bids set at incremental costs will not be sustainable over time. Bids based on the average costs (or the average heat rate) can be sustained. They may not appear in a standard run of a production simulation model, but they are much more likely to appear in a single-part bidding system.

B.4. The Northwest Power Planning Council

The NPPC has used a wide variety of models, many in support of the *Northwest Conservation and Electric Power Plan* (NPPC, 1991, 1996). Their most relevant report estimates future revenues if energy generation from the federal resources in the northwest were sold into a competitive market (NPPC, 1998). To develop market price forecasts, the NPPC relied principally on a proprietary model called Aurora. Aurora assumes that the competitive electricity market price will rise to the point of the variable cost of the last generating unit needed to meet demand. (Demand was represented by hourly demand in 12 regions of the WSCC, and a single demand growth rate was applied to all 12 regions. There are approximately 2000 generating units in the WSCC, and Aurora requires cost and operating specifications for each unit.)

Aurora builds a least-cost dispatch for the WSCC including transmission constraints, losses, wheeling costs and unit start-up costs in the dispatch. The market clearing price is then determined by observing the cost of meeting an incremental increase in the demand in each area. The hourly prices are aggregated to an average price over the entire year that would prevail under average hydro conditions. The average price for delivery in the northwest was forecasted at just under 20 \$/mwh in 1998. The first 5 yr of the forecast are dominated by the size of the surplus in generation within the WSCC and the price of natural gas.

Now, what about the long-term forecasts? Aurora calculates both long-term and short-term prices based on the least-cost dispatch. The “long-term” results appear automatically when the model performs an internal calculation of the investment in new generating capacity. The model assumes that investors have perfect foresight of market clearing prices in the future, and they put this knowledge to use as they contemplate building generating capacity. Thus, the long-run prices forecasts will “tend to settle at the price of the most cost-effective combination of new resources” (NPPC, 1998, p. 21). In most forecasts, the CCCTs make up the bulk (if not all) of the new resource additions, so the market clearing price ends up at the fully allocated cost of a new CCCT. With this reasoning, the NPPC’s medium forecast settles down to just under 25 \$/mwh (expressed in constant 1997 \$). With no changes in external factors, the price would remain at this value year after year after year.

The NPPC forecasts show essentially the same results as the CEC — the market price will follow a relatively smooth upward path and come into dynamic equilibrium at the fully allocated cost of a new CCCT. With this view, any volatility in the market clearing prices would be attributed to external factors such as variations in the price of natural gas or variations the amount of hydroelectric generation. This may be the viewpoint underpinning the CEC forecast, but the NPPC have a broader view of the future. They are aware that Aurora’s algorithm for adding generating capacity is simply not capable of shedding light on the possibility of a construction cycle. They know that a “perfect foresight” algorithm is a convenient assumption to allow the computer program to employ optimization techniques to arrive at one pattern of capacity additions in the future. They acknowledge that investors do not have perfect foresight and they warn that the Aurora forecasts “are likely to result in a smoother pattern of future prices than the actual market is likely to achieve” (NPPC, 1998, p. 25). They believe that the real electricity market is likely to experience price cycles of several years duration.

Acronyms and terminology

Aurora	proprietary production simulation model used by the NPPC
CCCT	combined cycle combustion turbine
CEC	California Energy Commission
CEGB	Central Electricity Generating Board (in the UK)
CfD	Contracts for differences (in the UK)
CTC	competitive transition charge (in California)
LOLP	loss of load probability
IOU	investor owned utility
IPP	independent power producer
NGC	National Grid Company (in the UK)
NPPC	Northwest Power Planning Council

Px	power exchange (in California)
SMP	system marginal price
UK	United Kingdom
VLL	value of loss of load

Units of measurement

Energy Energy is measured in kwh (kilo-watt-hour) or in aMW in this paper. An aMW is the same as 8760 mwh or 8,760,000 kwh. It is the energy generated by 1 MW of power operating over the 8760 h in a year.

Price of energy Energy prices are measured in mills/kwh or in \$/mwh in this paper. There are 1000 mills in a \$, and 1000 kwh in a mwh, so these prices are used interchangeably.

References

- Achi Z., Hausen, J., Nick, A., Pfeffer, J.-L., Verhaeghe, P. 1998. Managing capacity in basic materials. *Business Dynamics: Overcoming the Limits to Growth*, the McKinsey Quarterly Anthologies. McKinsey & Company, 55 East 52nd St, New York, NY 10022.
- Backus, G., Baylis, S., 1996. The dynamics of U.S. electric utility deregulation. Report to the office of Utility Technology. US Department of Energy, Policy Assessment Corporation, Denver, Colorado.
- Baxter, L., Hirst, E. 1995. Estimating potential stranded commitments for US investor-owned electric utilities. Oak Ridge National Laboratory Report ORNL/CON-406, January 1995.
- Bunn, D. (1994). Evaluating the effects of privatizing electricity. *Journal of the Operational Research Society* 45(4).
- Bunn, D., Day, C., Vlahos, K. 1997. Analysis of latent market power in the electricity pool of England and Wales. Working paper of the Energy Markets Group, London Business School, Regents Park, London NW1 4SA, UK, August 1997.
- Bunn, D., Larsen, E. 1992. Sensitivity of reserve margin to factors influencing investment behaviour in the electricity market of England and Wales. *Energy Policy* May.
- Cambridge Energy Research Associates (CERA) 1997. *North American Electric Power Watch*. Summer 1997.
- Coyle, G., 1977. *Management System Dynamics*. Wiley, New York.
- California Public Utilities Commission (CPUC) 1999. *Electric Restructuring Resource Notebook*. 505 Van Ness Ave, San Francisco, CA 94102-3298.
- Dickens, D., 1998. *Electricity Markets Principles and Practice*. St. Clements Services, Inc.
- California Energy Commissions, (ER) 1996. 1996 Electricity Report, November 1997, report P300-97-001.
- Farney, K., 1997. Lost margins, hidden capacity & high deliverability presentation to POWERGEN'97, Dallas, TX, December 10, 1997.
- Ford, A., 1990. Estimating the impact of efficiency standards on the uncertainty of the northwest electric system. *Operations Research* 38 (4).
- Ford, A., 1997. System dynamics and the electric power industry. *System Dynamics Review* 13 (1).
- Ford, A., 1999. *Modeling the Environment*. Island Press.
- Forrester, J., 1961. *Industrial Dynamics*. Pegasus Communications, Waltham, MA.
- Green, R., Newbery, D., 1992. Competition in the British electricity spot market. *Journal of Political Economy* 100 (5).
- Grow, R., 1996a. Incentives for new market entrants and the role of demand bidding staff testimony for California Energy Commission Proceedings on ER'96, June 18, 1996.
- Grow, R., 1996b. Playing the pool: can everybody win? *Public Utilities Fortnightly* March 1, 1996.
- Hastings, A., 1997. *Population Biology: Concepts and Models*. Springer, New York.
- Helm, D. (Ed.), 1994. *British Energy Policy in the 1990s: The Transition to the Competitive Market*. OXERA Press.
- Hirst, E., Hadley, S., Baxter, L. 1996. Factors that affect electric utility stranded commitments. Report ORNL/CON 432 of the Oak Ridge National Laboratory, July 1996.
- Hunt, S., Shuttleworth, G., 1996. *Competition and Choice in Electricity*. Wiley, New York.
- Joskow, P., 1997. Restructuring, competition and regulatory reform in the US electricity sector. *Journal of Economic Perspectives* 11 (3).
- Kahn, E., 1998. Price forecasting in spot markets. *Public Utilities Fortnightly* October 15, 1998.
- Krapels, E., Stagliano, V., 1996. Power in the commodity markets. *Public Utilities Fortnightly* May 15, 1996.
- Kaufmann, T., 1983. Aluminum industry prospects 1983–1993. *Mineral and Energy Resources* 26 (5).
- Klein, J., 1997. Interim staff market clearing price forecast for the California energy market. California Energy Commission, 1516 Ninth St, Sacramento, CA 95814, December 10, 1997.
- Littlechild, S., Director General OFFER, 1994. Transition to competition in electricity. In: Helm, D., (Ed.) *British Energy Policy in the 1990s*.
- Loehr, G., 1998. Ten myths about electric deregulation. *Public Utilities Fortnightly* April 15, 1998.
- Lomi, A., Larsen, E., 1999. Learning without experience: understanding the strategic implications of deregulation and competition in the electricity industry. *European Management Journal*, forthcoming.
- Meadows, D., 1970. *Dynamics of Commodity Production Cycles*. Pegasus Communications, Waltham, MA.
- McCullough, R., 1998. Can electricity markets work without capacity prices? *Public Utilities Fortnightly* March 15, 1998.
- Michaels, R., 1997. MW gamble: the missing market for capacity. *The Electricity Journal* December 1997.
- Moore, T., 1999. Merchant plants drive market competition. *EPRI Journal* 24 (2).
- Newbery, D., 1995. Power markets and market power. *The Energy Journal* 16 (3).
- Northwest Power Planning Council, (NPPC) 1998. Analysis of the Bonneville power administration's potential future costs and market revenues. 851 SW Sixth Ave, Portland, OR 97204.
- O'Donnell, A., 1996. Exceptions to the rule: bypassing the California transition charge. *Public Utilities Fortnightly*, Nov 15, 1996.
- Office of Electricity Regulation (OFFER) 1998. *Review of Electricity Trading Arrangements*. Hagley House, Birmingham B16 8QG, UK.
- Oregonian, 1995. Reynolds rides the roller coaster: ex-workers wait. *Oregonian*, April 27, 1995.
- Patterson, W., 1997. Electricity: liberal futures. Working Paper #2 on "Transforming Electricity" from the Energy and Environmental Programme of the Royal Institute of International Affairs. Chatham House, London.
- Ramos, A., Ventosa, M., Rivier, M., 1998. Modelling competition in electric energy markets by equilibrium constraints. *Utilities Policy* 7 (4).
- Richard, D., Lavinson, M., 1996. Something for everyone: the politics of California's new law on electric restructuring. *Public Utilities Fortnightly*, November 15, 1996.
- Rose, J., Mann, C., 1995. Unbundling the electric capacity price in a deregulated commodity market. *Public Utilities Fortnightly*, December 1995.
- Schuler, J. Jr., 1996. Generation: big or small? *Public Utilities Fortnightly* September 15, 1996.

- Sterman, J., 1999. *Business Dynamics*. McGraw-Hill, New York, forthcoming.
- Thurston, C., 1999. Merchant power: promise or reality? *Public Utilities* fortnightly January 1, 1999.
- Vlahos, K., 1998. The electricity markets microworld. Educational Document Series ED-0008-A, available from Professor Kiriakos Vlahos, The System Dynamics Group. London Business School.
- Wolak, F., Patrick, R., 1996. The impact of rules and market structure on the price determination process in the England and Wales electricity market. Working paper from Frank Wolak, Department of Economics, Stanford University.